Arrays 6

15 April 2026 16:42

arr = [1, 2, 3, 4, 5, 6, 7]

k = 3

After 1st rotation

[7, 1, 2, 3, 4, 5, 6]

$O(n^2) \rightarrow$!

$n \rightarrow$ arr. length

$n = 10^5$

$n^2 = 10^{10}$

10^8 → threshold!

Constraint

$< 10^4$

$< 10^5$

$< 10^8$

$O(n^3)$

$O(n^2 \log n)$

$O(n^2) \rightarrow$ ✗

$O(n \log n) \rightarrow$ max

$O(n)$

Inc. order

$< 10^9$ | $O(n)$ | $O(\log n)$

- Given → arr
  - sorted.

arr = $\overset{i}{\longleftrightarrow}$ $\overset{j}{\longleftrightarrow}$

$$\begin{array}{cccccc} [1, & 3, & 6, & 8, & 9, & 10] \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ i & i & i & j & j & j \end{array}$$

, target = 14

$1 + 10 = 11$

$3 + 10 = 13$

$6 + 10 = 16 > \text{target } (14)$

$6 + 9 = 15 > \text{target}$

$6 + 8 = 14 == \text{target}$

Ans→

```
int i=0, j= nums.length-1;

while (i<j) {
    if (nums[i] +num[j] == target)
        return {i+1, j+1};
    else if (nums[i] + nums[j] > target)
        j--;
    else i++;
return {-1, -1};
```